MIL-N-62623(AT)
30 March 1988
SUPERSEDING
(see 6.4)

MILITARY SPECIFICATION

NICKEL BASE ALLOY, SHEET, CORROSION- AND HEAT-RESISTANT

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sheet stock of a corrosion- and heat-resistant nickel base alloy.

2. APPLICABLE DOCUMENTS

2.1 Government document.

2.1.1 Standard. The following standard forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of this document shall be that listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARD
MILITARY

MIL-STD-45662 - Calibration Systems Requirements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 9630
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(Copies of standards required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	- Tension Testing of Metallic Materials.
ASTM E112	- Determining Average Grain Size.
ASTM E139	- Conducting Creep, Creep-Rupture, and Stress-Rupture Tests of Metallic Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 2262	- Tolerances, Nickel, Nickel Alloy, and Cobalt Alloy Sheet, Strip, and Plate.
AMS 2269	- Chemical Check Analysis Limits, Wrought Nickel Alloys and Cobalt Alloys.
AMS 2371	- Quality Assurance Sampling of Corrosion and Heat Resistant Steels and Alloys Wrought Products Except Forgings and Forging Stock.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish material which shall be subjected to first article inspection (see 4.4). The first article inspection sample, properly marked with identifying information, shall be representative of the material to be furnished to the Government. All subsequent material delivered to the Government shall conform to this sample in all of its pertinent physical and performance attributes.

3.2 Character or quality.

3.2.1 Condition. Unless otherwise specified (see 6.2), the material shall be delivered cold rolled and bright annealed (see 4.6.1).

3.2.2 Quality. The material shall be uniform in quality and condition, clean, sound, and free from foreign materials and from internal and external imperfections detrimental to fabrication or performance of parts (see 4.6.2).

3.3 Product characteristics.

3.3.1 Thickness. The material shall be not greater than 0.020 inch [0.51 millimeter (mm)] thick (see 4.6.2).

3.3.1.1 Tolerances. Unless otherwise specified (see 6.2), tolerances shall conform to AMS 2262 (see 4.6.2).

3.4 Chemical and mechanical properties.

3.4.1 Chemical composition. The chemical composition of the material shall conform to the percentages by weight of each element specified in table I (see 4.6.3).

TABLE I. Chemical composition.

Element	Percentage by weight	
	Minimum	Maximum
Carbon	--	0.10
Manganese	--	0.50
Silicon	--	0.50
Phosphorus	--	0.015
Sulfur	--	0.015
Chromium	20.00	23.00
Molybdenum	8.00	10.00
Columbium + Tantalum	3.15	4.15

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TABLE I. Chemical composition - Continued.

Element	Percentage by weight	
	Minimum	Maximum
Iron	--	5.00
Cobalt 1/	--	1.00
Titanium	Trace	0.40
Aluminum	Trace	0.40
Nickel	Remainder	

1/ Determination not required for routine acceptance.

3.4.1.1 Composition variations. Variations of the chemical composition shall be in accordance with AMS 2269 (see 4.6.3.1).

3.4.2 Tensile properties. The tensile properties of the material shall be as follows (see 4.6.4):

a. Ultimate strength: Not less than 105,000 pounds per square inch (psi) [724 megapascals (MPa)].
b. Yield strength at 0.2 percent (%) offset: 48,000 to 57,000 psi (331 to 393 MPa).
c. Elongation in 2 inches: Not less than 35%.

3.4.3 Stress rupture. The material shall not fail in less than 23 hours and elongation shall be not less than 15% in 2 inches, when maintained at a temperature of 1500 ± 5 degrees Fahrenheit (°F) [816 ± 3 degrees Celsius (°C)] while an axial load of 16,500 psi (114 MPa) is continuously applied (see 4.6.5).

3.4.4 Bending. The material shall withstand, without cracking, bending at room temperature through an angle of 180 degrees around a diameter equal to the nominal thickness of the material, with the axis of the bend parallel to the direction of rolling (see 4.6.6).

3.4.5 Grain size. The grain size of the material, when measured in accordance with ASTM E112, shall be ASTM #5 or finer (see 4.6.7).

3.4.6 Surface oxidation. Unless otherwise specified (see 6.2), there shall be no evidence of surface oxidation present on the material (see 4.6.8).

3.5 Identification marking. When identification marking is performed directly on the material, the marking fluid shall have no deleterious effect on the material and shall be sufficiently stable to withstand normal handling. The marking fluid shall be capable of being removed in a hot alkaline solution. Unless otherwise specified (see 6.2), the sheet shall be marked on one face as follows (see 4.6.2):

a. The rows of characters shall be spaced not over 3 feet (914 mm) apart.

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b. The characters shall be of a size to be clearly legible.
c. Marking shall include this specification number, revision letter, heat number, manufacturer's identification, and nominal thickness.

3.5.1 Flat sheet. Flat sheet over 6 inches (152 mm) in width shall be marked in lengthwise rows of characters. The rows of characters shall be spaced not more than 6 inches (152 mm) apart and alternately staggered.

3.5.2 Coiled sheet. Coiled sheet shall be marked near the outside end of the coil. The inside end of the coil shall also be marked, or shall have a tag or label attached and marked with the same information.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2.1 Specimen orientation. Tensile test specimens taken from widths 9 inches (229 mm) and over shall be taken with the axis perpendicular to the direction of rolling. For widths less than 9 inches (229 mm), tensile test specimens shall be taken with the axis of the tensile specimen parallel to the direction of rolling.

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4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F (23 ± 10°C).
b. Barometric pressure 28.5 (+2, -3) inches mercury. [725 (+50, -75) mm]
c. Relative humidity 50 ± 30%.

4.3 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   (1) Examination (see 4.5.2).
   (2) Acceptance tests (see 4.5.3).

4.4 First article inspection. The Government shall select sufficient sample material (see 6.2) produced under the production contract for first article inspection. The first article sample shall be inspected as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply material that is fully representative of that inspected as a first article sample. Any changes or deviation of the production material from the first article sample shall be subject to the approval of the contracting officer.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Acceptance tests
Condition	3.2.1	4.6.1	X		
Defects (see table III	3.3.2 thru 3.3.1.1 and 3.5 thru 3.5.2	4.6.2	X	X	
Chemical composition	3.4.1	4.6.3	X		X
Composition variation	3.4.1.1	4.6.3.1	X		X
Tensile properties	3.4.2	4.6.4	X		X
Stress rupture	3.4.3	4.6.5	X		X
Bending	3.4.4	4.6.6	X		X
Grain size	3.4.5	4.6.7	X		X
Surface oxidation	3.4.6	4.6.8	X		X

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4.4.1 Failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all material already produced prior to completion of the first article inspection is similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies shall be corrected by the contractor. The Government will not accept material until first article inspection is completed to the satisfaction of the Government.

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the material of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling of material. Unless otherwise specified (see 6.2), the material shall be sampled in accordance with AMS 2371.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of % defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Quality, not as specified (see 3.2.2).	Visual
102	Thickness, out of tolerance (see 3.3.1 and 3.3.1.1).	SIE 1/
Minor	AQL 2.5% Defective	
201	Marking, improper (see 3.5 thru 3.5.2).	Visual

1/ SIE = Standard Inspection Equipment.

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4.5.3 Acceptance tests. Each sample selected in accordance with 4.5.1.2 shall be subjected to the acceptance tests specified in table II.

4.5.3.1 Failure. Failure of any material to meet any acceptance test shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Condition. Conformance to 3.2.1 shall be determined by inspection of contractor records providing proof or certification that the condition of the material is cold rolled and bright annealed. Applicable records shall include specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2.2 through 3.3.1.1 and 3.5 through 3.5.2 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Chemical composition. To determine conformance to 3.4.1, the material shall be analyzed.

4.6.3.1 Composition variations. To determine conformance to 3.4.1.1, the material shall be analyzed in accordance with AMS 2269.

4.6.4 Tensile properties. To determine conformance to 3.4.2, the material shall be tested in accordance with ASTM E8.

4.6.5 Stress rupture. To determine conformance to 3.4.3, the material shall be tested in accordance with ASTM E139.

4.6.5.1 Increased loading. The stress rupture test may be conducted at a load higher than 16,500 psi (114 MPa), but the load shall not be changed while the test is in process. The time to rupture and the elongation shall remain the same.

4.6.6 Bending. To determine conformance to 3.4.4, the material shall be bent as specified and examined for any evidence of cracking.

4.6.7 Grain size. To determine conformance to 3.4.5, the material shall be tested in accordance with ASTM E112.

4.6.8 Surface oxidation. To determine conformance to 3.4.6, the material shall be examined for any evidence of surface oxidation.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. This material is intended primarily for parts requiring good corrosion- and oxidation-resistance at temperatures up to approximately 2000°F (1093°C), and where such parts may require welding.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of material (length, width, thickness).
c. If first article inspection is not required (see 3.1).
d. If the material shall be delivered in a condition other than as specified (see 3.2.1).
e. If tolerances shall be other than as specified (see 3.3.1.1).
f. If surface oxidation shall be other than as specified (see 3.4.6).
g. If marking shall be other than as specified (see 3.5).
h. If responsibility for inspection shall be other than as specified (see 4.1).
i. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
j. If inspection conditions shall be other than as specified (see 4.2).
k. First article inspection sample size (see 4.4).
l. If quality conformance inspection sampling shall be other than as specified (see 4.5.1.2).
m. Selection of applicable level and packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Corrosion-resistant
Heat-resistant
Nickel base alloy
Sheet

6.4 Supersession data. This military specification supersedes AVCO Lycoming Division specification no. M3622C, dated 26 April 1982.

Custodian:
Army - AT

Preparing activity:
Army - AT

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5. PROBLEM AREAS

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